



MATH140: Integration and Antiderivatives

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Antiderivatives — power rule

1. Find the antiderivative: integral of $2x^1 dx$.

$$\int 2x^1 dx$$

Answer: _____

2. Marginal cost is $MC(x) = 2x^2$. Find the total cost function $TC(x) = \text{integral of } MC dx$.

$$\int 2x^2 dx$$

Answer: _____

3. Find the antiderivative: integral of $2x^4 dx$.

$$\int 2x^4 dx$$

Answer: _____

4. Marginal cost is $MC(x) = 7x^1$. Find the total cost function $TC(x) = \text{integral of } MC dx$.

$$\int 7x^1 dx$$

Answer: _____

5. Find the antiderivative: integral of $8x^3 dx$.

$$\int 8x^3 dx$$

Answer: _____



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6. Marginal cost is $MC(x) = 5x^3$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 5x^3 \, dx$$

Answer: _____

7. Find the antiderivative: integral of $3x^4 \, dx$.

$$\int 3x^4 \, dx$$

Answer: _____

8. Marginal cost is $MC(x) = 3x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 3x^1 \, dx$$

Answer: _____

9. Find the antiderivative: integral of $4x^2 \, dx$.

$$\int 4x^2 \, dx$$

Answer: _____

10. Marginal cost is $MC(x) = 5x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 5x^1 \, dx$$

Answer: _____

11. Find the antiderivative: integral of $3x^1 \, dx$.

$$\int 3x^1 \, dx$$

Answer: _____

12. Marginal cost is $MC(x) = 9x^2$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 9x^2 \, dx$$

Answer: _____

13. Find the antiderivative: integral of $2x^3 \, dx$.

$$\int 2x^3 \, dx$$

Answer: _____



14. Marginal cost is $MC(x) = 3x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 3x^1 \, dx$$

Answer: _____

15. Find the antiderivative: integral of $2x^3 \, dx$.

$$\int 2x^3 \, dx$$

Answer: _____

16. Marginal cost is $MC(x) = 7x^3$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 7x^3 \, dx$$

Answer: _____

Definite integrals

17. Evaluate the definite integral: integral from 2 to 2 of $x^2 \, dx$.

$$\int_2^2 x^2 \, dx$$

Answer: _____

18. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^2 \, dx$$

Answer: _____

19. Evaluate the definite integral: integral from 1 to 3 of $x^3 \, dx$.

$$\int_1^3 x^3 \, dx$$

Answer: _____

20. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^1 \, dx$$

Answer: _____



21. Evaluate the definite integral: integral from 0 to 5 of $x^2 dx$.

$$\int_0^5 x^2 dx$$

Answer: _____

22. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^1 dx$$

Answer: _____

23. Evaluate the definite integral: integral from 0 to 5 of $x^3 dx$.

$$\int_0^5 x^3 dx$$

Answer: _____

24. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 3$ units.

$$\int_1^3 x^1 dx$$

Answer: _____

25. Evaluate the definite integral: integral from 1 to 3 of $x^2 dx$.

$$\int_1^3 x^2 dx$$

Answer: _____

26. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 0$ to $x = 4$ units.

$$\int_0^4 x^2 dx$$

Answer: _____

27. Evaluate the definite integral: integral from 0 to 5 of $x^2 dx$.

$$\int_0^5 x^2 dx$$

Answer: _____

28. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 1$ to $x = 5$ units.

$$\int_1^5 x^2 dx$$

Answer: _____



29. Evaluate the definite integral: integral from 1 to 2 of $x^3 dx$.

$$\int_1^2 x^3 dx$$

Answer: _____

30. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 0$ to $x = 4$ units.

$$\int_0^4 x^1 dx$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Antiderivatives — power rule, Definite integrals. All answers verified by independent computation.

Solutions



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Antiderivatives — power rule

1. Find the antiderivative: integral of $2x^1 dx$.

$$\int 2x^1 dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 1x^2 + C$.

Answer: $\frac{2}{2}x^2 + C \Rightarrow \text{coeff} = 1$

2. Marginal cost is $MC(x) = 2x^2$. Find the total cost function $TC(x) = \text{integral of } MC dx$.

$$\int 2x^2 dx$$

→ $TC(x) = \text{integral of } 2x^2 dx = \frac{2}{3}x^3 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{2}{3}x^3 + C \Rightarrow \text{coeff} = 2/3$

3. Find the antiderivative: integral of $2x^4 dx$.

$$\int 2x^4 dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= \frac{2}{5}x^5 + C$.

Answer: $\frac{2}{5}x^5 + C \Rightarrow \text{coeff} = 2/5$

4. Marginal cost is $MC(x) = 7x^1$. Find the total cost function $TC(x) = \text{integral of } MC dx$.

$$\int 7x^1 dx$$

→ $TC(x) = \text{integral of } 7x^1 dx = \frac{7}{2}x^2 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{7}{2}x^2 + C \Rightarrow \text{coeff} = 7/2$

5. Find the antiderivative: integral of $8x^3 dx$.

$$\int 8x^3 dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 2x^4 + C$.

Answer: $\frac{8}{4}x^4 + C \Rightarrow \text{coeff} = 2$



6. Marginal cost is $MC(x) = 5x^3$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 5x^3 \, dx$$

→ $TC(x) = \text{integral of } 5x^3 \, dx = 5/4x^4 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{5}{4}x^4 + C \Rightarrow \text{coeff} = 5/4$

7. Find the antiderivative: integral of $3x^4 \, dx$.

$$\int 3x^4 \, dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 3/5x^5 + C$.

Answer: $\frac{3}{5}x^5 + C \Rightarrow \text{coeff} = 3/5$

8. Marginal cost is $MC(x) = 3x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 3x^1 \, dx$$

→ $TC(x) = \text{integral of } 3x^1 \, dx = 3/2x^2 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{3}{2}x^2 + C \Rightarrow \text{coeff} = 3/2$

9. Find the antiderivative: integral of $4x^2 \, dx$.

$$\int 4x^2 \, dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 4/3x^3 + C$.

Answer: $\frac{4}{3}x^3 + C \Rightarrow \text{coeff} = 4/3$

10. Marginal cost is $MC(x) = 5x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 5x^1 \, dx$$

→ $TC(x) = \text{integral of } 5x^1 \, dx = 5/2x^2 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{5}{2}x^2 + C \Rightarrow \text{coeff} = 5/2$

11. Find the antiderivative: integral of $3x^1 \, dx$.

$$\int 3x^1 \, dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 3/2x^2 + C$.

Answer: $\frac{3}{2}x^2 + C \Rightarrow \text{coeff} = 3/2$



12. Marginal cost is $MC(x) = 9x^2$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 9x^2 \, dx$$

→ $TC(x) = \text{integral of } 9x^2 \, dx = 3x^3 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{9}{3}x^3 + C \Rightarrow \text{coeff} = 3$

13. Find the antiderivative: integral of $2x^3 \, dx$.

$$\int 2x^3 \, dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 1/2x^4 + C$.

Answer: $\frac{2}{4}x^4 + C \Rightarrow \text{coeff} = 1/2$

14. Marginal cost is $MC(x) = 3x^1$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 3x^1 \, dx$$

→ $TC(x) = \text{integral of } 3x^1 \, dx = 3/2x^2 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{3}{2}x^2 + C \Rightarrow \text{coeff} = 3/2$

15. Find the antiderivative: integral of $2x^3 \, dx$.

$$\int 2x^3 \, dx$$

→ Power rule for integration: $(a/(n+1))x^{(n+1)} + C$.

→ $= 1/2x^4 + C$.

Answer: $\frac{2}{4}x^4 + C \Rightarrow \text{coeff} = 1/2$

16. Marginal cost is $MC(x) = 7x^3$. Find the total cost function $TC(x) = \text{integral of } MC \, dx$.

$$\int 7x^3 \, dx$$

→ $TC(x) = \text{integral of } 7x^3 \, dx = 7/4x^4 + C$.

→ The constant C represents the fixed cost.

Answer: $\frac{7}{4}x^4 + C \Rightarrow \text{coeff} = 7/4$



Definite integrals

17. Evaluate the definite integral: integral from 2 to 2 of x^2 dx.

$$\int_2^2 x^2 dx$$

$$\rightarrow = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}.$$

$$\rightarrow = (\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}.$$

Answer: $\frac{8-8}{3} = 0$

18. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^2 dx$$

$$\rightarrow TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^n dx = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}.$$

$$\rightarrow = (\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}.$$

Answer: $\frac{216-1}{3} = 215/3$

19. Evaluate the definite integral: integral from 1 to 3 of x^3 dx.

$$\int_1^3 x^3 dx$$

$$\rightarrow = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}.$$

$$\rightarrow = (\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}.$$

Answer: $\frac{81-1}{4} = 20$

20. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^1 dx$$

$$\rightarrow TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^n dx = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}.$$

$$\rightarrow = (\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}.$$

Answer: $\frac{36-1}{2} = 35/2$

21. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

$$\rightarrow = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}.$$

$$\rightarrow = (\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}.$$

Answer: $\frac{125-0}{3} = 125/3$



22. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 6$ units.

$$\int_1^6 x^1 dx$$

→ $TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^{\{n\}} dx = [x^{\{n+1\}}/\{n+1\}] \text{ from } \{lo\} \text{ to } \{hi\}.$

→ $= (\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}.$

Answer: $\frac{36 - 1}{2} = 35/2$

23. Evaluate the definite integral: integral from 0 to 5 of $x^3 dx$.

$$\int_0^5 x^3 dx$$

→ $= [x^{\{n+1\}}/\{n+1\}] \text{ from } \{lo\} \text{ to } \{hi\}.$

→ $= (\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}.$

Answer: $\frac{625 - 0}{4} = 625/4$

24. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 1$ to $x = 3$ units.

$$\int_1^3 x^1 dx$$

→ $TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^{\{n\}} dx = [x^{\{n+1\}}/\{n+1\}] \text{ from } \{lo\} \text{ to } \{hi\}.$

→ $= (\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}.$

Answer: $\frac{9 - 1}{2} = 4$

25. Evaluate the definite integral: integral from 1 to 3 of $x^2 dx$.

$$\int_1^3 x^2 dx$$

→ $= [x^{\{n+1\}}/\{n+1\}] \text{ from } \{lo\} \text{ to } \{hi\}.$

→ $= (\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}.$

Answer: $\frac{27 - 1}{3} = 26/3$

26. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 0$ to $x = 4$ units.

$$\int_0^4 x^2 dx$$

→ $TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^{\{n\}} dx = [x^{\{n+1\}}/\{n+1\}] \text{ from } \{lo\} \text{ to } \{hi\}.$

→ $= (\{hi_pow\} - \{lo_pow\}) / \{n+1\} = \{answer_defint210\}.$

Answer: $\frac{64 - 0}{3} = 64/3$



27. Evaluate the definite integral: integral from 0 to 5 of x^2 dx.

$$\int_0^5 x^2 dx$$

→ = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{125 - 0}{3} = 125/3$

28. A marginal revenue function is $MR(x) = x^2$. Find total revenue from $x = 1$ to $x = 5$ units.

$$\int_1^5 x^2 dx$$

→ $TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^n dx = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{125 - 1}{3} = 124/3$

29. Evaluate the definite integral: integral from 1 to 2 of x^3 dx.

$$\int_1^2 x^3 dx$$

→ = $[x^{n+1}/(n+1)]$ from {lo} to {hi}.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{16 - 1}{4} = 15/4$

30. A marginal revenue function is $MR(x) = x^1$. Find total revenue from $x = 0$ to $x = 4$ units.

$$\int_0^4 x^1 dx$$

→ $TR = \text{integral from } \{lo\} \text{ to } \{hi\} \text{ of } x^n dx = [x^{n+1}/(n+1)] \text{ from } \{lo\} \text{ to } \{hi\}$.

→ = $(\{hi_pow\} - \{lo_pow\}) / (n+1) = \{answer_defint210\}$.

Answer: $\frac{16 - 0}{2} = 8$

